EAB 770 Problem Set 3 Key

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(emmeans)
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```
library(car)
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':
```

```
    some
```

```
library(MASS)
```

```
Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select
```

```
library(VGAM)
```

```
Warning: package 'VGAM' was built under R version 4.6.1

Loading required package: stats4
Loading required package: splines
```

```
library(survival)
```

1 Problem 1

CHD.csv includes data from a cardiovascular study on residents from the town of Framingham, Massachusetts. The researchers intend to pinpoint the most relevant risk factors of the disease. The response variable is TenYearCHD, which is the ten-year risk of coronary heart disease (CHD). A value TenYearCHD=1 means that there is a risk of CHD.

Test if there is sufficient evidence of an association between the ten-year risk of coronary heart disease and diabetes and prevalent hypertension accounting for age, number of cigarettes smoked daily, and sex. Interpret the resulting odds ratios in the context of the problem. You will be graded based on the interpretation and completeness of the results.

Solution

```
chd <- read.csv("CHD.csv")
glimpse(chd)
```

```
Rows: 4,238
```

```
Columns: 16
$ male            <int> 1, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 1, 0, ~
$ age             <int> 39, 46, 48, 61, 46, 43, 63, 45, 52, 43, 50, 43, 46, 41~
$ education       <int> 4, 2, 1, 3, 3, 2, 1, 2, 1, 1, 1, 2, 1, 3, 2, 2, 3, 2, ~
$ currentSmoker   <int> 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 1, 0, 1, 1, 1, 1, ~
$ cigsPerDay      <int> 0, 0, 20, 30, 23, 0, 0, 20, 0, 30, 0, 0, 15, 0, 9, 20,~
$ BPMeds          <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, ~
$ prevalentStroke <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ prevalentHyp    <int> 0, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, ~
$ diabetes        <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ totChol         <int> 195, 250, 245, 225, 285, 228, 205, 313, 260, 225, 254,~
$ sysBP           <dbl> 106.0, 121.0, 127.5, 150.0, 130.0, 180.0, 138.0, 100.0~
$ diaBP           <dbl> 70.0, 81.0, 80.0, 95.0, 84.0, 110.0, 71.0, 71.0, 89.0,~
$ BMI             <dbl> 26.97, 28.73, 25.34, 28.58, 23.10, 30.30, 33.11, 21.68~
$ heartRate       <int> 80, 95, 75, 65, 85, 77, 60, 79, 76, 93, 75, 72, 98, 65~
$ glucose         <int> 77, 76, 70, 103, 85, 99, 85, 78, 79, 88, 76, 61, 64, 8~
$ TenYearCHD      <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, ~
```

We use logistic regression to model the ten-year risk of coronary heart disease.

```
mod1 <- glm(TenYearCHD~diabetes + prevalentHyp + age + cigsPerDay + male,
            data=chd,
            family="binomial")

summary(mod1)
```

```

Call:
glm(formula = TenYearCHD ~ diabetes + prevalentHyp + age + cigsPerDay +
    male, family = "binomial", data = chd)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -6.115296   0.317903 -19.236  < 2e-16 ***
diabetes      0.868458   0.213607   4.066 4.79e-05 ***
prevalentHyp  0.675276   0.093137   7.250 4.16e-13 ***
age           0.071872   0.005769  12.459  < 2e-16 ***
cigsPerDay    0.021280   0.003835   5.549 2.87e-08 ***
male          0.421413   0.095907   4.394 1.11e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3595.1  on 4208  degrees of freedom
Residual deviance: 3242.1  on 4203  degrees of freedom
  (29 observations deleted due to missingness)
AIC: 3254.1

Number of Fisher Scoring iterations: 5
```

```
Anova(mod1)
```

```
Analysis of Deviance Table (Type II tests)

Response: TenYearCHD
             LR Chisq Df Pr(>Chisq)
diabetes       15.306  1  9.142e-05 ***
prevalentHyp   51.925  1  5.768e-13 ***
age           164.574  1  < 2.2e-16 ***
cigsPerDay     30.102  1  4.098e-08 ***
male           19.284  1  1.126e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There is strong evidence of associations with the ten-year risk of CHD for those with diabetes (χ^2_1=15.3,p<0.001), and prevalent hypertension (χ^2_1=51.925, p<0.0001) after accounting for age, number of cigarettes smoked per day, and sex.
To calculate odds ratios, we use `emmeans`.

```
emm_diabetes <- emmeans(mod1,~diabetes,type="response",infer=T)
contrast(emm_diabetes, method="revpairwise",infer=T)
```

```
 contrast              odds.ratio    SE  df asymp.LCL asymp.UCL null z.ratio
 diabetes1 / diabetes0       2.38 0.509 Inf      1.57      3.62    1   4.066
 p.value
 <0.0001

Results are averaged over the levels of: prevalentHyp, male
Confidence level used: 0.95
Intervals are back-transformed from the log odds ratio scale
Tests are performed on the log odds ratio scale
```

The odds of reporting a ten-year risk of CHD for patients with diabetes is 2.38 (95% CI:1.57-3.62) times that of patients without diabetes after accounting for age, number of cigarettes smoked per day, and sex.

```
emm_hyp <- emmeans(mod1,~prevalentHyp,type="response",infer=T)
contrast(emm_hyp, method="revpairwise",infer=T)
```

```
 contrast                    odds.ratio    SE  df asymp.LCL asymp.UCL null
 prevalentHyp1 / prevalentHyp0       1.96 0.183 Inf      1.64      2.36    1
 z.ratio p.value
   7.250 <0.0001

Results are averaged over the levels of: diabetes, male
Confidence level used: 0.95
Intervals are back-transformed from the log odds ratio scale
Tests are performed on the log odds ratio scale
```

The odds of reporting a ten-year risk of CHD for patients with hypertension is 1.96 (95% CI:1.64-2.36) times that of patients without hypertension after accounting for age, number of cigarettes smoked per day, and sex.

2 Problem 2

The file clinics.csv features data from a multi-clinic trial that tests Treatment 1 to a placebo treatment (Treatment 0) and a positive control (Treatment 2). The response recorded was the severity of symptoms after treatment (Slight, Moderate, Severe). Use the proportional odds model to test for a treatment effect after accounting for the multiple hospital sites. Calculate and interpret the relevant odds ratios.

```
clinics <- read.csv('clinics.csv')
glimpse(clinics)
```

```
Rows: 45
Columns: 4
$ hospital <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3~
$ trt      <int> 0, 0, 0, 1, 1, 1, 2, 2, 2, 0, 0, 0, 1, 1, 1, 2, 2, 2, 0, 0, 0~
$ response <chr> "slight", "moderate", "severe", "slight", "moderate", "severe~
$ count    <int> 1, 4, 23, 2, 7, 23, 4, 7, 18, 3, 6, 11, 5, 9, 15, 9, 5, 11, 0~
```

Solution

We implement the proportional odds model in R using `polr()` from the `MASS` package.

```
clinics <- clinics |>
  uncount(weights=count)

clinics <- clinics |>
  mutate(response = factor(response,levels=c("slight","moderate","severe")),
         trt = as.factor(trt),
         hospital = as.factor(hospital))

mod_po <- polr(response~trt + hospital,
               data=clinics)

poTest(mod_po)
```

```
Tests for Proportional Odds
polr(formula = response ~ trt + hospital, data = clinics)

           b[polr] b[>slight] b[>moderate] Chisquare df Pr(>Chisq)
Overall                                         3.71  6       0.72
trt1      -0.15230   -0.24076     -0.13316      0.07  1       0.80
trt2      -1.36809   -1.33999     -1.36889      0.01  1       0.94
hospital2 -1.13452   -1.28686     -1.00762      0.38  1       0.54
hospital3 -1.18289   -1.15076     -1.19134      0.01  1       0.93
hospital4 -0.61436   -0.33660     -0.69771      0.49  1       0.49
hospital5 -0.36988    0.04388     -0.49909      0.87  1       0.35
```

The proportional odds assumption holds for both treatment (p=0.77) and hospital (p=0.12).

```
Anova(mod_po)
```

```
Analysis of Deviance Table (Type II tests)

Response: response
         LR Chisq Df Pr(>Chisq)
trt        35.693  2  1.776e-08 ***
hospital   19.005  4  0.0007842 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There is strong evidence of a treatment effect (χ^2_1=35.7, p<0.001). We now investigate the relevant odds ratios.

```
emm_treatment <- emmeans(mod_po,~trt)
```

```
Re-fitting to get Hessian
```

```
contrast(emm_treatment,method="pairwise",infer=T) |>
  as.data.frame() |>
  mutate(OR = exp(estimate),
         OR.LCL = exp(asymp.LCL),
         OR.UCL = exp(asymp.UCL))
```

```
     contrast  estimate        SE  df  asymp.LCL asymp.UCL   z.ratio
1 trt0 - trt1 0.1522981 0.2656112 Inf -0.4702150 0.7748112 0.5733873
2 trt0 - trt2 1.3680911 0.2632344 Inf  0.7511486 1.9850337 5.1972357
3 trt1 - trt2 1.2157930 0.2485904 Inf  0.6331716 1.7984145 4.8907486
       p.value       OR    OR.LCL   OR.UCL
1 8.343213e-01 1.164507 0.6248679 2.170182
2 6.056121e-07 3.927846 2.1194329 7.279293
3 3.003026e-06 3.372968 1.8835751 6.040063
```

There is no evidence of a difference between the treatment and the negative control (OR=1.16, 95%CI: 0.62-2.17, p=0.83). However, there is strong evidence of a difference between the treatment and the positive control (OR=3.37, 95%CI: 1.88-6.04, p<0.0001). The odds of reporting a more severe response in the treatment is 3.37 times that of the positive control and 1.16 times that of the negative control

3 Problem 3

The presence of sprouted or diseased kernels in wheat can reduce the value of a wheat producer's entire crop. A study was conducted examining numerous physical properties of kernels—hardness, size, weight, and moisture content—measured on a sample of wheat kernels from two different classes of wheat, hard red winter (hrw) and soft red winter (srw). Each kernel's condition was also classified as "Healthy," "Sprout," or "Scab" by human visual inspection. The data is included in the file wheat.csv. Use the generalized logits model to create a main effects model that tests the effect of the class and physical properties of the wheat kernels on the kernel's condition.

```r
wheat <- read.csv("wheat.csv")
glimpse(wheat)
```

```
Rows: 275
Columns: 6
$ class     <chr> "hrw", "hrw", "hrw", "hrw", "hrw", "hrw", "hrw", "hrw", "hrw~
$ size      <dbl> 2.30274, 2.72573, 2.51246, 2.27164, 2.35478, 2.49132, 3.0613~
$ weight    <dbl> 24.6480, 33.2985, 31.7580, 32.7060, 26.0700, 33.2985, 36.498~
$ moisture  <dbl> 12.01538, 12.17396, 11.87949, 12.11407, 12.06487, 12.18577, ~
$ hardness  <chr> "Hard", "Hard", "Hard", "Hard", "Hard", "Hard", "Hard", "Har~
$ condition <chr> "Healthy", "Healthy", "Healthy", "Healthy", "Healthy", "Heal~
```

Solution

```r
wheat <- wheat |>
  mutate(condition = factor(condition, levels = c("Sprout","Healthy","Scab")),
         hardness = factor(hardness,levels= c("Soft","Hard")))
modv <- vglm(condition~hardness + class + size + weight + moisture,
             data=wheat,
             family=multinomial(parallel=FALSE, refLevel=1))

summary(modv)
```

```
Call:
vglm(formula = condition ~ hardness + class + size + weight +
    moisture, family = multinomial(parallel = FALSE, refLevel = 1),
    data = wheat)

Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept):1  -0.43195    1.29184  -0.334  0.73810
(Intercept):2   4.65189    1.55260   2.996  0.00273 **
hardnessHard:1  1.04399    0.37082   2.815  0.00487 **
hardnessHard:2  0.22341    0.47125   0.474  0.63544
classsrw:1      0.09762    0.41752   0.234  0.81514
classsrw:2     -0.56439    0.52471  -1.076  0.28209
size:1         -0.58901    0.47311  -1.245  0.21314
size:2          0.26787    0.62871   0.426  0.67006
weight:1        0.02831    0.03087   0.917  0.35924
weight:2       -0.26656    0.05053  -5.275 1.33e-07 ***
```

```
moisture:1      0.01125    0.09826   0.115  0.90881
moisture:2      0.13417    0.12396   1.082  0.27907
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,2]/mu[,1]), log(mu[,3]/mu[,1])

Residual deviance: 478.6337 on 538 degrees of freedom

Log-likelihood: -239.3169 on 538 degrees of freedom

Number of Fisher scoring iterations: 6


Reference group is level  1  of the response
```

```
Anova(modv)
```

```
Analysis of Deviance Table (Type II tests)

Response: condition
         Df   Chisq Pr(>Chisq)
hardness  2  8.0758    0.01763 *
class     2  1.7460    0.41770
size      2  2.3923    0.30236
weight    2 34.3495  3.476e-08 ***
moisture  2  1.3011    0.52176
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
exp(coef(modv))
```

```
 (Intercept):1  (Intercept):2 hardnessHard:1 hardnessHard:2     classsrw:1
     0.6492407    104.7833472      2.8405414      1.2503323      1.1025400
    classsrw:2         size:1         size:2       weight:1       weight:2
     0.5687078      0.5548743      1.3071730      1.0287101      0.7660078
    moisture:1     moisture:2
     1.0113183      1.1435915
```

```
exp(confint(modv))
```

```
                    2.5 %       97.5 %
(Intercept):1  0.05161697    8.1661793
(Intercept):2  4.99710529 2197.1820134
hardnessHard:1 1.37327307    5.8755068
hardnessHard:2 0.49647448    3.1488647
classsrw:1     0.48640997    2.4991150
classsrw:2     0.20335677    1.5904489
size:1         0.21952497    1.4025076
size:2         0.38122228    4.4821654
weight:1       0.96830720    1.0928809
weight:2       0.69377506    0.8457611
moisture:1     0.83415414    1.2261100
moisture:2     0.89692594    1.4580931
```

There is strong evidence of an association with hardness (χ^2_2=8.08, p=0.017) and weight (χ^2_2=34.3,p<0.001).
The odds of observing healthy vs. sprouted kernels for hard kernels is estimated to be 2.841 (95%CI: 1.37, 5.88) times that of soft kernels. The odds of observing scabbed vs. sprouted kernels for hard kernels is 1.25 (95%CI: 0.50,3.15) times that of soft kernels. The odds ratio of observing healthy vs. sprouted kernels for the kernels that are heavier by one unit in weight is 1.03 (95%CI: 0.97, 1.09). The odds ratio of observing scabbed vs. sprouted kernels for the kernels that are heavier by one unit in weight is 0.77 (95%CI: 0.69,0.85).

4 Problem 4

The infert data set in R, also loaded in as infert.csv for SAS users, includes data from a matched case-control study that investigated infertility after spontaneous and induced abortion. The strata are given by the variable "stratum". The number of prior induced abortions are encoded as the "induced" variable (0=zero induced abortions, 1 = 1 induced abortion, 2=2+ induced abortions), while the number of prior spontaneous abortions are encoded as the "spontaneous variable" (0=zero spontaneous abortions, 1 = 1 spontaneous abortion, 2=2+ spontaneous abortions). The "case" variable is the outcome variable, where 1 = case, 0 = control.

```
glimpse(infert)
```

```
Rows: 248
Columns: 8
$ education      <fct> 0-5yrs, 0-5yrs, 0-5yrs, 0-5yrs, 6-11yrs, 6-11yrs, 6-
11y~
```

```
$ age            <dbl> 26, 42, 39, 34, 35, 36, 23, 32, 21, 28, 29, 37, 31, 29,~
$ parity         <dbl> 6, 1, 6, 4, 3, 4, 1, 2, 1, 2, 2, 4, 1, 3, 2, 2, 5, 1, 3~
$ induced        <dbl> 1, 1, 2, 2, 1, 2, 0, 0, 0, 0, 1, 2, 1, 2, 1, 2, 2, 0, 2~
$ case           <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ spontaneous    <dbl> 2, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 1, 1~
$ stratum        <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, ~
$ pooled.stratum <dbl> 3, 1, 4, 2, 32, 36, 6, 22, 5, 19, 20, 37, 9, 29, 21, 18~
```

```
infert <- infert |>
  mutate(across(c(parity,induced,spontaneous,stratum,pooled.stratum),~as.factor(.x)))
```

4.1 Problem 4a

Use the forward selection method with conditional logistic regression to find the most appropriate model. Treat "stratum" as the stratification variable. Use the base and full models with predictors shown below:

- Base model: spontaneous+ induced
- Full model: spontaneous + induced + age + Education + parity

Solution

```
base_model <- clogit(case ~spontaneous+ induced + strata(stratum),data=infert,model=TRUE)
full_model <- clogit(case~ spontaneous + induced + age + education + parity + strata(stratu
selected_model <- stepAIC(base_model,list(upper=full_model),direction="forward")
```

```
Start:  AIC=136.35
Surv(rep(1, 248L), case) ~ spontaneous + induced + strata(stratum)

       Df    AIC
<none>    136.35
```

b. Using your model in (a), interpret the results of the conditional logistic regression. (HINT: induced and spontaneous are technically categorical variables. Ensure that this is accounted for in your analysis.)

Solution

```
Anova(selected_model)
```

```
Warning in Anova.coxph(selected_model): LR tests not supported for models with clusters or
```

```
 Wald tests substituted

skipping term strata(stratum)

Analysis of Deviance Table (Type II tests)

Response: Surv(rep(1, 248L), case)
            Df  Chisq Pr(>Chisq)
spontaneous  2 31.855   1.21e-07 ***
induced      2 15.150  0.0005131 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There is strong evidence of an association between infertility and prior spontaneous abortions (χ^2_2=31.86,p<0.001) and prior induced abortions (χ^2_2=15.15, p=0.0005).